SRINIVAS UNIVERSITY INSTITUTE OF ENGINEERING AND TECHNOLOGY

MODULE 5

NATURAL LANGUAGE PROCESSING

GROUP TASK

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GROUP TASK

Chatbot Build & Demo: Groups create a simple rule-based chatbot (e.g., with Dialogflow). Present its logic, user interactions, and improvements

Introduction

A chatbot is a software application that interacts with users using natural language. It uses Natural Language Processing (NLP) techniques to understand user input and provide meaningful responses. Chatbots are widely used in customer service, e-commerce, education, healthcare, and banking systems.

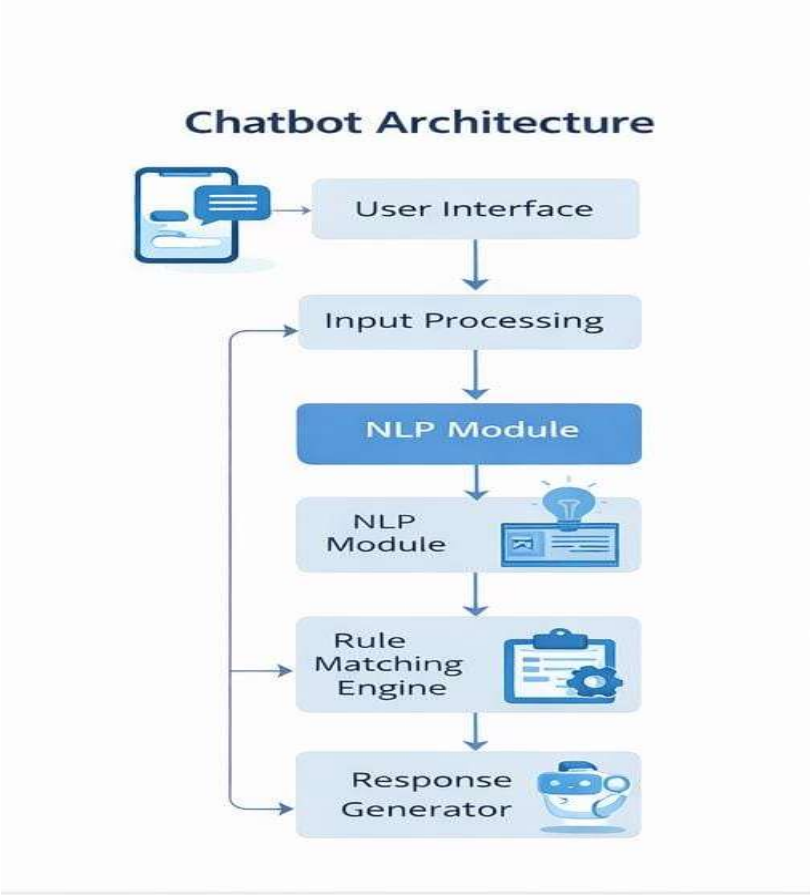
In this group task, a rule-based chatbot is designed and implemented. The chatbot responds to user queries based on predefined rules. It identifies keywords in user input and generates appropriate responses. This chatbot demonstrates how NLP techniques like tokenization, intent detection, and keyword matching are used.

The chatbot system follows a simple flow:
User input → NLP processing → Intent detection → Rule matching → Response generation → Output

Chatbot System Architecture

1. **User Interface**
   This is the place where the user interacts with the chatbot. The user enters a message using text or voice. The chatbot receives the input from the user.
2. **Input Processing**
   In this stage, the user input is cleaned and prepared.
   Unnecessary symbols are removed and the sentence is converted into a proper format for processing.

3. **NLP Module**
   The NLP module understands the user message using Natural Language Processing techniques. It performs tokenization, intent detection, and keyword extraction.
4. **Rule Matching Engine**
   The rule matching engine compares the extracted keywords with predefined rules stored in the chatbot. It finds the best matching rule.
5. **Response Generator**
   After matching the rule, the response generator creates the appropriate reply based on the selected rule.
6. **Output Display**
   Finally, the chatbot displays the generated response to the user. The user can read the reply and continue the conversation.



Chatbot Design Flow

Step 1: User enters message
The user types a message or question to the chatbot. This message is the input for the chatbot system.
Example: "I want to buy a mobile"

Step 2: Text preprocessing
The chatbot cleans the input text by removing special characters, converting text to lowercase, and preparing it for processing.

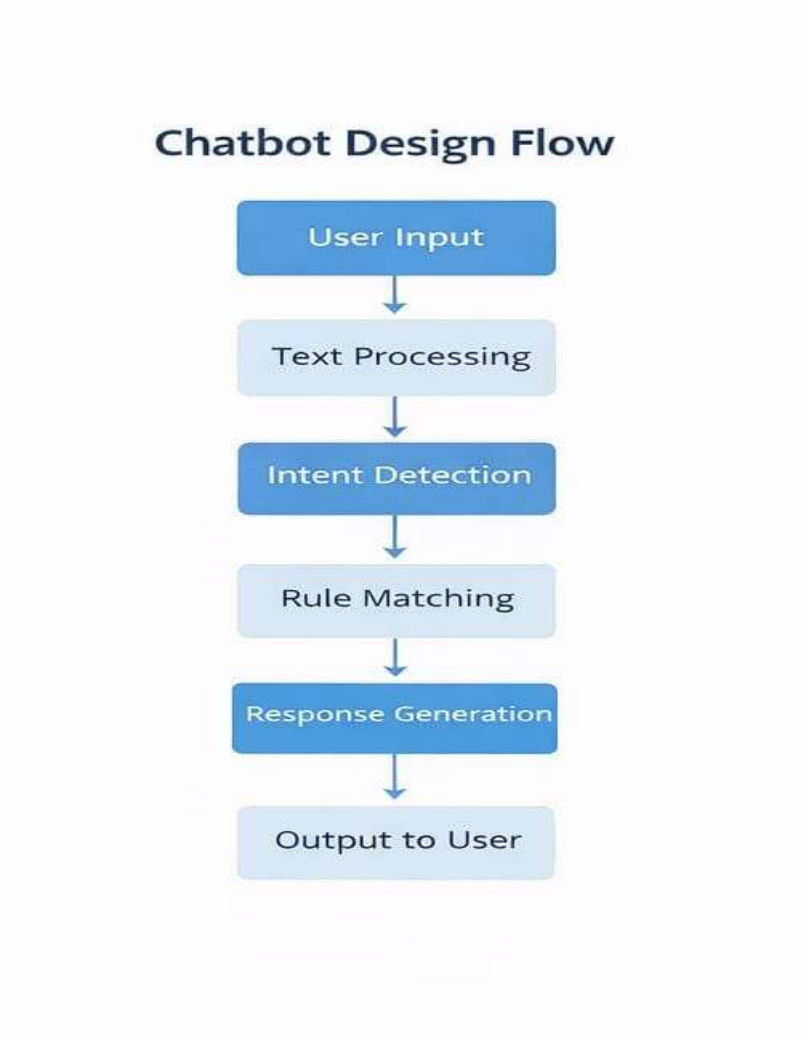
Step 3: Tokenization
The sentence is divided into individual words called tokens. This helps the chatbot understand each word clearly.
Example:
"I want to buy mobile" → I | want | buy | mobile

Step 4: Intent detection
The chatbot identifies the user's intention from the tokens. It understands what the user wants.
Example: Intent detected → Purchase product

Step 5: Rule matching
The chatbot compares the detected intent or keywords with predefined rules stored in the system. It finds the matching rule.

Step 6: Response generation
After finding the correct rule, the chatbot generates the appropriate reply based on the matched rule.

Step 7: Display output
Finally, the chatbot displays the generated response to the user. The user can then continue the conversation.



Chatbot Rules and Logic

- The chatbot follows **rule-based logic** to respond to user queries.
- Each user input is checked for **keywords** like hello, price, buy, thanks.
- The chatbot compares the input with **predefined rules** stored in the system.
- When a keyword matches, the chatbot **selects the corresponding response**.
- The response is generated automatically without human involvement.
- This process is called **keyword matching** in rule-based chatbot.
- The chatbot only responds to **predefined inputs**.

- If no rule matches, chatbot shows **default response** like "Please try again".
- This method is **simple and fast**.
- It is suitable for **basic conversation systems**.

EXAMPLE:

Advantages

- Simple to design
- Easy implementation
- Fast response time

- Low cost
- No training required
- Works for basic queries
- Good for small applications

Limitations

- Cannot understand complex sentences
- Limited knowledge
- No learning capability
- Cannot understand context deeply
- Works only for predefined rules

Applications

Customer support chatbot
College enquiry chatbot
Library chatbot
Shopping chatbot
Banking chatbot
Help desk chatbot

Result

The chatbot was successfully designed and implemented. The system accepted user input and responded using predefined rules. The chatbot identified keywords and generated correct responses. The interaction between user and chatbot was smooth.

The chatbot worked correctly for greeting, product enquiry, purchase request, and confirmation messages. The chatbot successfully simulated real-time conversation.

Conclusion

In this group task, a rule-based chatbot was designed using NLP techniques. The chatbot processed user input, identified keywords, and generated appropriate responses. The chatbot demonstrated how NLP is used in conversational AI systems.

This task helped understand chatbot architecture, rule-based logic, and NLP processing. The chatbot successfully responded to user queries and completed conversation flow. Hence, NLP plays an important role in building chatbot systems.